WASTE-TO-ENERGY DOWNDRAFT GASIFIER

System Design & Technical Specification Engineering Report

Project: Group 6 Final Project Report | **Target Capacity:** 1 kW Electrical Output

This technical document details the engineering specifications, operational architecture, and subsystem design for the 1 kW Waste-to-Energy Downdraft Gasifier system. Designed to process densified biomass (paper, cardboard, sawdust, and agricultural residues), the system converts solid waste into clean synthetic gas (syngas) to drive an internal combustion engine generator for localized off-grid power production.

1. GAS ENGINE / GENERATOR SUBSYSTEM

The gas engine/generator unit serves as the primary mechanical-to-electrical energy conversion interface in the Waste-to-Energy plant.

Engine & Electrical Specifications

- **Prime Mover:** Commercial spark-ignition petrol genset converted for syngas operation, with a minimum engine displacement of **178 cm³** (typically rated 2.5 – 3.0 kVA on petrol).
- **Net Electrical Output: 1.0 kW** continuous power output.
- **Air-to-Fuel Ratio Control:** Configured for gaseous fuel intake with an optimized volumetric mixing ratio of approximately **1.1 : 1 (Air to Syngas)** at the engine intake manifold.
- **Volumetric Intake Stream:** Total mixture flow rate of **11.2 Nm³/h** (comprising 5.3 Nm³/h clean syngas and 5.9 Nm³/h primary/secondary intake air).
- **System Efficiency:** Overall fuel-to-electricity conversion efficiency of **10.5%** (typical benchmark for micro-scale gasification-generator units).

2. MONITORING SYSTEM & INSTRUMENTATION

To ensure optimal thermochemical conversion, tar management, and operational stability, the gasifier is integrated with a comprehensive sensor suite and telemetry interface.

- **Drying & Pyrolysis Zone (T1)**
- Temperature
- 200 °C – 450 °C
- Monitors feedstock thermal degradation & drying rate
- **Oxidation / Throat Zone (T2)**
- Temperature
- 800 °C – 1000 °C
- Ensures complete cracking of heavy tars
- **Reduction Bed (T3)**
- Temperature
- 600 °C – 800 °C
- Maintains endothermic CO and H_2 formation
- **Hot Raw Syngas Outlet (#3)**
- Temperature & Tar Load
- ~450 °C (carrying dust & residual tar)
- Monitors cyclone inlet thermal conditions
- **Fine Fabric Filter**
- Differential Pressure (dP)
- Monitored dP limit

- Detects filter particulate loading & clogging
- **Engine Intake** / **Ambient Zone**
- Carbon Monoxide (CO)
- Continuous safety threshold tracking
- Triggers visual/audible alarm & emergency shutdown

Stream / Location	Measured Parameter	Operating Range / Target Standard	Control Function

3. SAFETY SYSTEMS & HAZARD MITIGATION

Producer gas contains approximately **20% Carbon Monoxide (CO)**, which is extremely toxic, odourless, colorless, and explosive when mixed with air. Strict engineering controls and protocols are mandated:

> **CRITICAL SAFETY MANDATE:** The gasifier plant must operate exclusively in open-air outdoor environments with trained supervisory personnel present at all times. Indoor operation is strictly prohibited.

- **Emergency Gas Flare Subsystem:** Integrated burn-off flare with a thermal rating of **6.7 kW thermal**. Used during start-up, pre-heating, and emergency engine shut-down to safely combust raw or excess syngas before environmental release.
- **Backfire & Explosion Prevention:** In-line **flame arrestors** and **safety check valves** are installed upstream of the gas buffer tank and engine intake to prevent flame propagation back into fuel lines.
- **Leak Detection & Purging Protocol:** Systematic pressure leak-test required across all flange joints and pipe connections prior to ignition; mandatory nitrogen/air purging procedures before cold start-ups.
- **Feedstock Restrictions:** Exclusively dry biomass pellets (6–10 mm, <15% moisture). Plastics (PET, PVC, PE) are strictly banned as they melt, clog the reduction bed, and produce highly toxic/dieldrin dioxin emissions.
- **Thermal & Hazardous Material Protection:** Mechanical guards fitted around hot surfaces (>60 °C), char ash pots, and tar water discharge collection units.

4. RESIDUE HANDLING & BY-PRODUCT MANAGEMENT

Gasification generates inorganic solid residues and liquid condensates that require systematic handling to prevent environmental contamination and operational degradation.

By-Product & Waste Stream Management

- **Solid Ash & Char Discharge:** Dry bottom ash and unreacted char accumulate at the hearth grate/ash pot at a rate of **0.1 kg/h**. Ash is periodically evacuated via an airtight grate mechanism and cooled prior to disposal or soil enrichment application.
- **Particulate & Fly Ash Separation:** High-efficiency cyclone separator traps heavy airborne particulates and fly ash immediately downstream of the gasifier reactor exit.
- **Tar & Aqueous Condensate Handling:** Hot syngas (~450 °C) is cooled to ~40 °C in a gas cooler (heat rejection duty: 0.7 kW) and passed through a water scrubber. Condensed tar-water and heavy oils drain directly into a dedicated **Condensate Tank** for safe batch neutralization and disposal.
- **Gas Conditioning & Buffer Storage:** Cleaned syngas passes through a fabric fine filter to eliminate micro-particles before entering a low-pressure gas buffer tank to smooth pressure oscillations prior to engine injection.

Design Basis Summary: Biomass Feed Input: 2.23 kg/h (as fed, 12% moisture) | Air Flow Rate: 2.6 Nm^3/h via 3 nozzles (3.1 mm dia.) | Syngas Production: 5.3 Nm^3/h (6.7 kW thermal output) | Reactor ID: 140 mm | Throat ID: 35 mm.